

A Appendix

Table provides detailed information on the SLMs used in this study, including model names, training data descriptions, and performance rankings on the Hugging Face Leaderboard.

Table 1: Top SLMs with HuggingFace Links and Dataset Types

Rank	Model Name	Link	Dataset
1	google/gemma-2-2b-jpn-it	https://huggingface.co/google/gemma-2-2b-jpn-it	General Text, Code, Mathematical/Logical Content
2	microsoft/phi-2	https://huggingface.co/microsoft/phi-2	General Text, Code, Mathematical/Logical Content
3	ibm-granite/granite-3.0-2b-base	https://huggingface.co/ibm-granite/granite-3.0-2b-base	General Text, Code, Mathematical/Logical Content
4	Qwen/Qwen2.5-1.5B	https://huggingface.co/Qwen/Qwen2.5-1.5B	General Text, Code, Mathematical/Logical Content
5	google/flan-t5-xl	https://huggingface.co/google/flan-t5-xl	General Text, Code, Mathematical/Logical Content
6	Qwen/Qwen2-1.5B	https://huggingface.co/Qwen/Qwen2-1.5B	General Text, Code, Mathematical/Logical Content
7	google/gemma-2-2b	https://huggingface.co/google/gemma-2-2b	General Text, Code, Mathematical/Logical Content
8	HuggingFaceTB/SmolLM2-1.7B	https://huggingface.co/HuggingFaceTB/SmolLM2-1.7B	General Text, Code, Mathematical/Logical Content
9	google/flan-t5-large	https://huggingface.co/google/flan-t5-large	General Text, Code, Mathematical/Logical Content
10	Qwen/Qwen1.5-1.8B	https://huggingface.co/Qwen/Qwen1.5-1.8B	General Text, Code
11	tensoropera/Fox-1-1.6B	https://huggingface.co/tensoropera/Fox-1-1.6B	General Text
12	google/gemma-2b	https://huggingface.co/google/gemma-2b	General Text, Code, Mathematical/Logical Content

13	stabilityai/stablelm-3b-4e1t	https://huggingface.co/stabilityai/stablelm-3b-4e1t	General Text, Code, Mathematical/Logical Content
14	Qwen/Qwen2-0.5B	https://huggingface.co/Qwen/Qwen2-0.5B	General Text, Code, Mathematical/Logical Content
15	microsoft/phi-1_5	https://huggingface.co/microsoft/phi-1_5	General Text, Code
16	google/codegemma-1.1-2b	https://huggingface.co/google/codegemma-1.1-2b	General Text, Code, Mathematical/Logical Content
17	google/recurrentgemma-2b	https://huggingface.co/google/recurrentgemma-2b	General Text, Code, Mathematical/Logical Content
18	HuggingFaceTB/SmolLM-135M	https://huggingface.co/HuggingFaceTB/SmolLM-135M	General Text, Code
19	openai-community/gpt2	https://huggingface.co/openai-community/gpt2	General Text
20	allenai/OLMo-1B-hf	https://huggingface.co/allenai/OLMo-1B-hf	General Text, Code, Mathematical/Logical Content
21	BEE-spoke-data/smol_llama-220M-GQA	https://huggingface.co/BEE-spoke-data/smol_llama-220M-GQA	General Text, Code, Mathematical/Logical Content
22	EleutherAI/gpt-neo-2.7B	https://huggingface.co/EleutherAI/gpt-neo-2.7B	General Text, Code, Mathematical/Logical Content
23	princeton-nlp/sheared-LLaMA-2.7B	https://huggingface.co/princeton-nlp/sheared-LLaMA-2.7B	General Text, Code
24	Qwen/Qwen2.5-0.5B	https://huggingface.co/Qwen/Qwen2.5-0.5B	General Text, Code, Mathematical/Logical Content
25	google/flan-t5-base	https://huggingface.co/google/flan-t5-base	General Text, Code, Mathematical/Logical Content
26	HuggingFaceTB/SmolLM-360M	https://huggingface.co/HuggingFaceTB/SmolLM-360M	General Text, Code
27	HuggingFaceTB/SmolLM2-360M	https://huggingface.co/HuggingFaceTB/SmolLM2-360M	General Text, Code, Mathematical/Logical Content

28	google/flan-t5-small	https://huggingface.co/google/flan-t5-small	General Text, Code, Mathematical/Logical Content
29	ibm-granite/granite-3.0-1b-a400m-base	https://huggingface.co/ibm-granite/granite-3.0-1b-a400m-base	General Text, Code, Mathematical/Logical Content
30	openai-community/gpt2-medium	https://huggingface.co/openai-community/gpt2-medium	General Text
31	EleutherAI/pythia-160m	https://huggingface.co/EleutherAI/pythia-160m	General Text, Code, Mathematical/Logical Content
32	HuggingFaceTB/SmolLM2-135M	https://huggingface.co/HuggingFaceTB/SmolLM2-135M	General Text, Code, Mathematical/Logical Content
33	microsoft/phi-1	https://huggingface.co/microsoft/phi-1	General Text, Code, Mathematical/Logical Content
34	princeton-nlp/sheared-LLaMA-1.3B	https://huggingface.co/princeton-nlp/sheared-LLaMA-1.3B	General Text, Code
35	openai-community/gpt2-large	https://huggingface.co/openai-community/gpt2-large	General Text
36	vonjack/MobileLLM-125M-HF	https://huggingface.co/vonjack/MobileLLM-125M-HF	General Text, Code, Mathematical/Logical Content
37	EleutherAI/pythia-2.8b	https://huggingface.co/EleutherAI/pythia-2.8b	General Text, Code, Mathematical/Logical Content
38	HuggingFaceTB/SmolLM-1.7B	https://huggingface.co/HuggingFaceTB/SmolLM-1.7B	General Text, Code, Mathematical/Logical Content
39	EleutherAI/gpt-neo-1.3B	https://huggingface.co/EleutherAI/gpt-neo-1.3B	General Text, Code, Mathematical/Logical Content
40	facebook/opt-1.3b	https://huggingface.co/facebook/opt-1.3b	General Text
41	stabilityai/stablelm-2-1_6b	https://huggingface.co/stabilityai/stablelm-2-1_6b	General Text, Code
42	amd/AMD-Llama-135m	https://huggingface.co/amd/AMD-Llama-135m	General Text, Code

43	Qwen/Owen1.5-0.5B	https://huggingface.co/Qwen/Owen1.5-0.5B	General Text, Code
44	EleutherAI/pythia-410m	https://huggingface.co/EleutherAI/pythia-410m	General Text, Code, Mathematical/Logical Content
45	keeeeenw/MicroLlama	https://huggingface.co/keeeeenw/MicroLlama	General Text
46	openai-community/gpt2-xl	https://huggingface.co/openai-community/gpt2-xl	General Text
47	TinyLlama/TinyLlama_v1.1	https://huggingface.co/TinyLlama/TinyLlama_v1.1	General Text
48	EleutherAI/gpt-neo-125m	https://huggingface.co/EleutherAI/gpt-neo-125m	General Text, Code, Mathematical/Logical Content
49	pints-ai/1.5-Pints-16K-v0.1	https://huggingface.co/pints-ai/1.5-Pints-16K-v0.1	General Text, Mathematical/Logical Content
50	M4-ai/TinyMistral-248M-v3	https://huggingface.co/M4-ai/TinyMistral-248M-v3	General Text, Mathematical/Logical Content
51	meta-llama/Llama-3.2-1B	https://huggingface.co/meta-llama/Llama-3.2-1B	General Text
52	bigscience/bloom-1b7	https://huggingface.co/bigscience/bloom-1b7	General Text
53	bigscience/bloom-1b1	https://huggingface.co/bigscience/bloom-1b1	General Text
54	BEE-spoke-data/smol_llama-101M-GQA	https://huggingface.co/BEE-spoke-data/smol_llama-101M-GQA	General Text, Code
55	distilbert/distilgpt2	https://huggingface.co/distilbert/distilgpt2	General Text, Code
56	pints-ai/1.5-Pints-2K-v0.1	https://huggingface.co/pints-ai/1.5-Pints-2K-v0.1	General Text, Mathematical/Logical Content
57	bigscience/bloom-560m	https://huggingface.co/bigscience/bloom-560m	General Text

58	instruction- pretrain/Instruct LM-500M	https:// huggingface.co/ instruction-pretrain/ InstructLM-500M	General Text, Mathemat- ical/Logical Content
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To evaluate training data diversity, we adopt a simple quantification method by assigning scores based on the types of content included in each dataset. The scoring criteria are as follows:

- General text (1 point): Includes standard textual content such as news articles, novels, and social media posts.
- Code (1 point): Includes programming code or technical content such as API documentation or coding tutorials.
- Math/reasoning (1 point): Includes mathematical reasoning, equations, physics problems, or scientific texts involving logical inference.

Table 7 summarizes the training data volume and parameter size of the SLMs used in this study. The model index corresponds to the performance rank listed in Table 6.

Table 2: SLM Training Data Volume and Parameter Sizes (Trillion Tokens & Billion Parameters)

Model Index	Data Volume (Trillion Tokens)	Parameter (Billion)
1	2	2.61
2	1.4	2.78
3	12	2.53
4	18	1.54
5	0.3	2.85
6	3	1.54
7	2	2.61
8	11	1.70
9	0.3	0.783
10	3	1.84
11	3	1.60
12	2	2.51
13	1	3.00
14	12	0.50
15	0.03	1.42
16	1	2.51
17	1	2.68
18	0.25	0.135
19	0.02	0.117
20	3	1.00
21	0.002	0.022
22	0.42	2.70

23	0.005	0.027
24	5.5	0.050
25	0.3	0.248
26	0.25	0.360
27	4	0.360
28	0.3	0.077
29	10	0.138
30	0.02	0.0345
31	0.3	0.160
32	2	0.135
33	0.054	0.142
34	0.005	0.013
35	0.02	0.0774
36	1	0.125
37	0.3	2.80
38	0.25	1.70
39	0.38	1.30
40	0.18	1.30
41	2	1.64
42	0.67	1.35
43	7	0.050
44	0.3	0.041
45	0.05	0.305
46	0.02	0.150
47	1.5	1.10
48	0.3	1.25
49	0.057	1.57
50	0.003	0.248
51	9	1.00
52	0.18	1.70
53	0.18	1.10
54	0.002	1.01
55	0.001	0.82
56	0.057	1.60
57	0.16	0.56
58	0.1	0.50

The SecuDevSLM framework examined 58 SLMs in this study, and the training dataset information for each model was collected based on their respective publishers as follows:

A.1 Google

(1) google/gemma-2-2b-jpn-it

- Language: Fine-tuned on Japanese.

- Web Documents: A rich and diverse collection of web texts ensures the model is exposed to a wide range of language styles, topics, and vocabulary, primarily in Japanese.
- Code: Exposure to code helps the model learn programming syntax and patterns, enhancing its ability to generate or understand code-related tasks.
- Mathematics: Training on mathematical texts supports learning in logical reasoning, symbolic representation, and solving mathematical problems.

(2) google/flan-t5-xl

- openai/gsm8k: High-quality, linguistically diverse elementary math word problems.
- cimec/lambada: Narrative passages extracted from a book corpus.
- deepmind/code_contests: Programming problem datasets.

(3) google/gemma-2-2b

- Web Documents: A diverse set of web texts ensures exposure to a broad range of language styles, topics, and vocabulary, primarily in English.
- Code: Code exposure enhances the model’s ability to learn programming languages and improve its performance in code generation or comprehension tasks.
- Mathematics: Mathematical content supports the development of logical reasoning and problem-solving abilities.

(4) google/flan-t5-large

- openai/gsm8k: High-quality, linguistically diverse elementary math problems.
- cimec/lambada: Narrative passages from a book corpus.
- deepmind/code_contests: Programming problems.

(5) google/gemma-2b

- Web Documents: A diverse web text collection enables the model to access a wide range of language styles and topics, mainly in English.
- Code: Code exposure helps the model learn programming syntax and structure.
- Mathematics: Training with math texts aids in logic, symbol manipulation, and solving math-related queries.

(6) google/codegemma-1.1-2b

- Open-source mathematical datasets
- Synthetic code
- Public code repositories

(7) google/recurrentgemma-2b

- Web Documents: Varied web text collections exposing the model to diverse linguistic styles and topics, mostly in English.
- Code: Code exposure enhances learning of programming syntax and tasks.
- Mathematics: Supports reasoning and problem-solving using mathematical texts.

(8) google/flan-t5-base

- openai/gsm8k: High-quality elementary math problems in various languages.
- cimec/lambada: Narratives from a book corpus.

- deepmind/code_contests: Programming challenge data.
- **(9) google/flan-t5-small**
- openai/gsm8k: Multilingual elementary math word problems.
- cimec/lambada: Book-based narrative segments.
- deepmind/code_contests: Programming problems.

A.2 Microsoft

(10) microsoft/phi-2

- NLP: Synthetic texts for natural language processing tasks.
- Code: Python subset from Stack v1.2.
- Q&A: Question-and-answer data from StackOverflow.
- Programming Contests: Code from code_contests programming competitions.
- Synthetic Textbook: Synthetic Python textbooks and exercises generated by gpt-3.5-turbo-0301.

(11) microsoft/phi-1.5

- NLP: Synthetic texts for natural language processing tasks.
- Code: Python subset from Stack v1.2.
- Q&A: Question-and-answer data from StackOverflow.
- Programming Contests: Code from code_contests programming competitions.
- Synthetic Textbook: Synthetic Python textbooks and exercises generated by gpt-3.5-turbo-0301.

(12) microsoft/phi-1

- Code: Python subset from Stack v1.2.
- Q&A: Question-and-answer data from StackOverflow.
- Programming Contests: Code from code_contests programming competitions.
- Synthetic Textbook: Synthetic Python textbooks and exercises generated by gpt-3.5-turbo-0301.

A.3 IBM

(13) ibm-granite/granite-3.0-2b-base

- Pretraining Stage 1: Data from academia, internet, enterprise domains (e.g., finance, law), and code.
- Pretraining Stage 2: Curated high-quality data from the same domains, including multilingual and instructional data.

(14) ibm-granite/granite-3.0-1b-a400m-base

- Pretraining Stage 1: Data from academia, internet, enterprise domains (e.g., finance, law), and code.
- Pretraining Stage 2: Curated high-quality data from the same domains, including multilingual and instructional data.

A.4 Qwen

(15) Qwen/Qwen2.5-1.5B

- Data: Public web documents, encyclopedias, books, and code.
- Focus: Natural language understanding, general question answering, coding, mathematics, scientific knowledge, reasoning, and multilingual ability.
- Strength: Significant enhancements in coding and mathematics.

(16) Qwen/Qwen2-1.5B

- Data: Public web documents, encyclopedias, books, and code.
- Focus: Natural language understanding, general question answering, coding, mathematics, scientific knowledge, reasoning, and multilingual ability.

(17) Qwen/Qwen1.5-1.8B

- Data: Public web documents, encyclopedias, books, and code.
- Strength: Notable improvement in conversational capabilities.

(18) Qwen/Qwen2-0.5B

- Data: Public web documents, encyclopedias, books, and code.
- Focus: Natural language understanding, general question answering, coding, mathematics, scientific knowledge, reasoning, and multilingual ability.

(19) Qwen/Qwen2.5-0.5B

- Data: Public web documents, encyclopedias, books, and code.
- Focus: Natural language understanding, general question answering, coding, mathematics, scientific knowledge, reasoning, and multilingual ability.
- Strength: Significant enhancements in coding and mathematics.

(20) Qwen/Qwen1.5-0.5B

- Data: Public web documents, encyclopedias, books, and code.
- Strength: Notable improvement in conversational capabilities.

A.5 HuggingFaceTB

(21) HuggingFaceTB/SmolLM2-1.7B

- FineWeb-Edu: Deduplicated educational web samples from FineWeb.
- DCLM: A standardized language corpus.
- The Stack: Code data.
- argilla/Synth-APIGen-v0.1: A set of function-related problems in mathematics, physics, and programming.

(22) HuggingFaceTB/SmolLM-135M

- Cosmopedia v2: Synthetic textbooks and stories generated by Mixtral.
- Python-Edu: Educational Python samples from The Stack.
- FineWeb-Edu: Deduplicated educational web samples from FineWeb.

(23) HuggingFaceTB/SmolLM-360M

- Cosmopedia v2: Synthetic textbooks and stories generated by Mixtral.
- Python-Edu: Educational Python samples from The Stack.
- FineWeb-Edu: Deduplicated educational web samples from FineWeb.

(24) HuggingFaceTB/SmolLM2-360M

- FineWeb-Edu: Deduplicated educational web samples from FineWeb.
- DCLM: A standardized language corpus.
- The Stack: Code data.
- argilla/Synth-APIGen-v0.1: A set of function-related problems in mathematics, physics, and programming.

(25) HuggingFaceTB/SmolLM2-135M

- FineWeb-Edu: Deduplicated educational web samples from FineWeb.
- DCLM: A standardized language corpus.
- The Stack: Code data.
- argilla/Synth-APIGen-v0.1: A set of function-related problems in mathematics, physics, and programming.

(26) HuggingFaceTB/smolLM-1.7B

- Cosmopedia v2: Synthetic textbooks and stories generated by Mixtral.
- Python-Edu: Educational Python samples from The Stack.
- FineWeb-Edu: Deduplicated educational web samples from FineWeb.

A.6 OpenAI**(27) openai-community/gpt2**

- English Text Corpus: Trained only on raw text to predict the next word in a sentence.
- Web Data: Covers a broad range of topics and domains to ensure diverse knowledge.

(28) openai-community/gpt2-medium

- English Text Corpus: Trained only on raw text to predict the next word in a sentence.
- Web Data: Covers a broad range of topics and domains to ensure diverse knowledge.

(29) openai-community/gpt2-large

- English Text Corpus: Trained only on raw text to predict the next word in a sentence.
- Web Data: Covers a broad range of topics and domains to ensure diverse knowledge.

(30) openai-community/gpt2-xl

- English Text Corpus: Trained only on raw text to predict the next word in a sentence.
- Web Data: Covers a broad range of topics and domains to ensure diverse knowledge.

A.7 EleutherAI**(31) EleutherAI/gpt-neo-2.7B**

- Pile Dataset: Includes Pile-CC (web), PubMed Central (medical), Books3, OpenWebText2, ArXiv (scientific), GitHub (code), FreeLaw (legal), Stack-Exchange (Q&A), USPTO Backgrounds (patents), PubMed Abstracts, Gutenberg (PG-19), OpenSubtitles, Wikipedia (en), DM Mathematics, Ubuntu IRC (chat logs), BookCorpus2, EuroParl (parliament records), HackerNews, YoutubeSubtitles, PhilPapers (philosophy), NIH ExPorter (research funding), Enron Emails.
- (32) EleutherAI/pythia-160m**
- Pile Dataset: Same sources as EleutherAI/gpt-neo-2.7B.
- (33) EleutherAI/pythia-2.8b**
- Pile Dataset: Same sources as EleutherAI/gpt-neo-2.7B.
- (34) EleutherAI/gpt-neo-1.3B**
- Pile Dataset: Same sources as EleutherAI/gpt-neo-2.7B.
- (35) EleutherAI/pythia-410m**
- Pile Dataset: Same sources as EleutherAI/gpt-neo-2.7B.
- (36) EleutherAI/gpt-neo-125m**
- Pile Dataset: Same sources as EleutherAI/gpt-neo-2.7B.

A.8 StabilityAI

- (37) stabilityai/stablelm-3b-4e1t**
- tiuuai/falcon-refinedweb: Web data based on CommonCrawl, including raw and processed web content.
- bigcode/starcoderdata: Code datasets.
- allenai/peS2o: Academic papers across biology, chemistry, engineering, computer science, physics, material science, mathematics, psychology, economics, political science, business, geology, sociology, geography, environmental science, art, history, and philosophy.
- (38) stabilityai/stablelm-2-1_6b**
- tiuuai/falcon-refinedweb: Web data based on CommonCrawl.
- bigcode/starcoderdata: Code datasets.
- uonlp/CulturaX: Multilingual web content.

A.9 Princeton NLP

- (39) princeton-nlp/sheared-LLaMA-2.7B**
- RedPajama-Data: News articles, academic papers, social media, technical documents, literature, and encyclopedia entries.
- (40) princeton-nlp/sheared-LLaMA-1.3B**
- RedPajama-Data: Same as (39).

A.10 TensorOpera

- (41) tensoropera/Fox-1-1.6B**

- 3 trillion web-scraped documents.

A.11 AllenAI

(42) allenai/OLMo-1B-hf

- allenai/dolma includes:
 - Dolma’s CC, Refined Web, C4: Web data
 - StarCoder: Code data
 - Reddit: Social media
 - Semantic Scholar (S2ORC & S2AG), arXiv: Academic papers
 - StackExchange: Q&A forums
 - Flan: Multi-task datasets
 - CC News: News articles
 - OpenWebMath, Algebraic Stack: Math data
 - Project Gutenberg: Books
 - MegaWika, Wikipedia & Wikibooks: Encyclopedia content

A.12 BEE-spoke-data

(43) BEE-spoke-data/smol_llama-220M-GQA

- JeanKaddour/minipile: A small Pile subset including medical, code, legal, news, subtitles, books, chats, emails.
- mattymchen/refinedweb-3m: Web content.
- pszemraj/simple_wikipedia_LM: NLP-focused data.

(44) BEE-spoke-data/smol_llama-101M-G0A

- JeanKaddour/minipile: Same as (43).
- mattymchen/refinedweb-3m: Web content.
- pszemraj/simple_wikipedia_LM: NLP-focused data.

A.13 Meta

(45) meta-llama/Llama-3.2-1B

- Multilingual text
- Public web data

(46) facebook/opt-1.3b

- BookCorpus: Over 10K unpublished books
- CC-Stories: Filtered story subset
- The Pile: Includes Pile-CC, OpenWebText2, USPTO, Gutenberg, OpenSubtitles, Wikipedia, DM Mathematics, HackerNews
- Reddit dataset
- CCNewsV2 dataset

A.14 AMD**(47) amd/AMD-Llama-135m**

- cerebras/SlimPajama-627B: News, academic papers, social media, technical docs, literature, encyclopedias
- manu/project_gutenberg: Large collection of ebooks

A.15 VonJack**(48) vonjack/MobileLLM-125M-HF**

- Public online data

A.16 Keeneenw**(49) keeneenw/MicroLlama**

- No code training
- cerebras/SlimPajama-627B: News, academic papers, social media, technical docs, literature, encyclopedias

A.17 TinyLlama**(50) TinyLlama/TinyLlama_v1.1**

- cerebras/SlimPajama-627B: News, academic papers, social media, technical docs, literature, encyclopedias

A.18 Pints-AI**(51) pints-ai/1.5-Pints-16K-v0.1**

- HuggingFaceH4/ultrachat_200k: Chat dialogues
- HuggingFaceH4/ultrafeedback_binarized: Prompt-feedback data
- meta-math/MetaMathQA: Math questions

(52) pints-ai/1.5-Pints-2K-v0.1

- HuggingFaceH4/ultrachat_200k: Chat dialogues
- HuggingFaceH4/ultrafeedback_binarized: Q&A data
- meta-math/MetaMathQA: Math questions

A.19 M4-AI**(53) M4-ai/TinyMistral-248M-v3**

- mlfoundations/dclm-baseline-1.0: Large-scale web data
- LLM360/TxT360: Math and physics problems
- Skylion007/openwebtext: Web text

A.20 BigScience

(54) bigscience/bloom-1b7

- Trained for text generation, information extraction, Q&A, and summarization.

(55) bigscience/bloom-1b1

- Trained for text generation, information extraction, Q&A, and summarization.

(56) bigscience/bloom-560m

- Trained for text generation, information extraction, Q&A, and summarization.

A.21 DistilBERT

(57) distilbert/distilgpt2

- Multilingual text
- Programming content

A.22 Instruction-Pretrain

(58) instruction-pretrain/InstructLM-500M

- tiuuue/falcon-refinedweb: Web content from CommonCrawl
- instruction-pretrain/general-instruction-augmented-corpora: Instruction-augmented web text
- instruction-pretrain/ft-instruction-synthesizer-collection: Mathematics data